

LAB-04

**LOUD ML PLATFORM
VIPARITIVE ANALYSIS**

GROUP-#04

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1. INTRODUCTION:

Machine learning has become a cornerstone of modern data-driven decision-making across industries. As organizations scale their ML initiatives, the choice of cloud platform plays a critical role in determining the speed, cost, and quality of model development and deployment. Three platforms have emerged as dominant options in the enterprise ML landscape: **AWS SageMaker**, **Google AutoML (Vertex AI)**, and **Azure ML Studio**.

This report provides a structured comparison of these three platforms through the lens of a real-world use case: healthcare insurance fraud detection. Rather than evaluating platforms in the abstract, we apply each to the same end-to-end ML pipeline from data ingestion through production monitoring and assess their relative strengths, limitations, and fit for different organizational contexts.

Actual implementation is not required for this assignment. The objective is exploratory: to develop a working understanding of how each platform approaches the full ML lifecycle, and to build the analytical vocabulary needed to evaluate cloud ML services in professional settings.

2. Case Selection: Healthcare Insurance Fraud Detection

Healthcare fraud costs the United States an estimated \$300 billion annually, placing enormous financial and operational strain on insurance providers, government programs and ultimately patients. Insurance companies require sophisticated, scalable ML systems capable of flagging suspicious claims before payment is processed by reducing losses, deterring fraudulent providers, and improving resource allocation.

Problem Statement

Given historical claims data including patient demographics, provider information, procedure codes (ICD/CPT), billing amounts, claim frequency, and historical payment patterns build a binary classification model to predict whether a new claim is **fraudulent (1)** or **legitimate (0)**.

Why This Use Case?

Healthcare insurance fraud detection presents a rich, multi-dimensional ML challenge that serves as an ideal benchmark for comparing cloud platforms:

- **Structured tabular data:** All three platforms handle tabular classification natively, enabling fair comparison.
- **Class imbalance:** Fraud is rare (typically 1–5% of claims), requiring specialized handling: SMOTE, class weighting, or oversampling.
- **Regulatory compliance:** HIPAA governance in healthcare makes model explainability, bias detection, and audit trails non-negotiable.
- **Real-time inference:** Claims scoring must occur before payment is released, demanding low-latency endpoints.
- **Data drift risk:** Fraud patterns evolve continuously, making ongoing monitoring and retraining essential.

ML Pipeline Overview

Stage	Description
Data Ingestion	Import structured claims data from data warehouses, CSV files, or streaming sources (e.g., Kafka).
Preprocessing	Handle missing values, encode categorical variables (ICD codes, provider IDs), normalize numeric features, and address class imbalance via SMOTE or class weighting.
Feature Engineering	Create derived features: claim frequency per provider, average billing amount per procedure, patient claim history ratios.
Model Training	Train classification models (XGBoost, Random Forest, Neural Networks). Tune hyperparameters and cross-validate.
Model Evaluation	Assess using Precision, Recall, F1-Score, AUC-ROC, and Confusion Matrix. Prioritize Recall to minimize missed fraud cases.
Deployment	Deploy a real-time scoring endpoint for incoming claims. Set decision threshold aligned with business risk tolerance.
Monitoring	Track data drift, model performance degradation, and prediction confidence over time. Trigger retraining as needed.

3. PLATFORM EXPLORATION:

AWS SageMaker:

AWS SageMaker is Amazon's fully managed end-to-end ML platform, tightly integrated with the broader AWS ecosystem. It offers the most comprehensive set of ML tools available on any single cloud, spanning data preparation, model training, hyperparameter optimization, deployment, and production monitoring, all within a unified interface backed by S3, Glue, CloudWatch and EventBridge.

Data Ingestion & Storage	Raw claims data is stored in Amazon S3 as CSV or Parquet. AWS Glue catalogs and performs ETL from relational sources (RDS, Redshift). SageMaker Data Wrangler provides a visual no-code data preparation interface.
Preprocessing & Feature Engineering	Data Wrangler offers 300+ built-in transformations for imputation, encoding, and normalization. SageMaker Processing Jobs run custom Python scripts (Scikit-learn, PySpark) at scale. SageMaker Feature Store provides a centralized, reusable feature repository with online and offline access.

Model Training	Built-in optimized XGBoost runs distributed natively — ideal for tabular fraud data. Script Mode supports TensorFlow, PyTorch, and custom containers. Automatic Model Tuning uses Bayesian search for efficient hyperparameter optimization. Class imbalance is addressed via <code>scale_pos_weight</code> in XGBoost or SMOTE in Processing Jobs.
Model Evaluation	SageMaker Experiments automatically tracks all training runs, metrics, and artifacts. SageMaker Clarify detects bias in predictions and provides SHAP-based feature explanations — critical for HIPAA compliance. Metrics are visualized via CloudWatch or exported to S3 for custom dashboards.
Deployment & Inference	Models are deployed to a Real-Time Endpoint (HTTPS REST) for synchronous claim scoring. Asynchronous Inference handles large batch claim files. Auto-scaling manages traffic spikes during peak claim periods. A/B testing via SageMaker Endpoints enables safe, gradual model rollout.
Monitoring	SageMaker Model Monitor automatically detects data drift, model quality degradation, and bias shifts. Amazon SNS alerts notify stakeholders when performance drops. Retraining pipelines are triggered automatically via SageMaker Pipelines and EventBridge.

Google AutoML (Vertex AI):

Google's Vertex AI consolidates AutoML and custom training under a single unified platform. AutoML Tables, the tabular data component, is designed to reduce the ML expertise barrier automating algorithm selection, preprocessing, and model evaluation. Vertex AI integrates deeply with BigQuery, making it the natural choice for data-warehouse-centric analytics teams already operating within the Google Cloud ecosystem.

Data Ingestion & Storage	Claims data is stored in Google Cloud Storage (GCS) as CSV or in BigQuery as a managed data warehouse. Vertex AI Datasets connects directly to BigQuery by enabling seamless ingestion without manual export. Dataflow (Apache Beam-based) handles large-scale serverless ETL pipelines.
Preprocessing & Feature Engineering	AutoML Tables automatically handles missing values, encoding, and feature normalization with minimal manual intervention. Vertex AI Feature Store provides a managed, shareable registry of ML features backed by BigQuery. Cloud Dataprep (Trifacta) offers visual wrangling for custom transformations.
Model Training	AutoML Tables trains and internally selects the best model testing ensemble methods and neural architecture search automatically. No algorithm selection is required from the user. For custom models, Vertex AI Custom Training supports pre-built or custom containers (TF, PyTorch, Scikit-learn). Vertex AI Vizier provides standalone hyperparameter

	optimization. Class imbalance is handled automatically through internal class weighting.
Model Evaluation	AutoML Tables produces an auto-generated Model Evaluation report containing confusion matrix, AUC-ROC, precision-recall curves at multiple thresholds, and feature importance scores. Vertex Explainable AI provides SHAP-based attributions per prediction which is valuable for regulatory audits. Experiment tracking is managed via Vertex AI Experiments with TensorBoard integration.
Deployment & Inference	Models are deployed to a Vertex AI Online Prediction Endpoint for real-time REST-based inference. Batch Prediction Jobs score thousands of claims in parallel using managed compute. Traffic splitting across model versions enables safe rollouts.
Monitoring	Vertex AI Model Monitoring tracks input feature distribution drift and prediction drift. Alert policies in Google Cloud Monitoring trigger retraining workflows automatically. Vertex AI Pipelines (Kubeflow-based) automates the full retraining and redeployment cycle.

Azure ML Studio:

Azure ML Studio is Microsoft's enterprise-grade ML platform, deeply integrated with the Azure data ecosystem (Blob Storage, Synapse, Data Factory, Power BI). Its standout feature for regulated industries is the Responsible AI Dashboard which is the most comprehensive fairness, explainability, and error analysis suite of the three platforms. Azure ML offers a dual interface: a visual Designer for non-engineers and a full Python SDK for data scientists, making it uniquely accessible to mixed-expertise teams.

Data Ingestion & Storage	Claims data is stored in Azure Blob Storage or Azure Data Lake Storage Gen2. Azure Data Factory and Synapse Analytics handle ETL pipelines. Azure ML Datastores and Datasets provide versioned, reproducible data references within the ML workspace.
Preprocessing & Feature Engineering	Azure ML Designer offers a visual drag-and-drop interface with pre-built modules for data cleaning, normalization, and feature selection. The Azure ML SDK (Python) supports custom preprocessing scripts on managed compute clusters. Azure ML Feature Store (preview) centralizes engineered features with offline/online serving. Synapse Pipelines handle complex, large-scale ETL before data reaches ML Studio.
Model Training	Automated ML (AutoML) evaluates dozens of algorithms and ensembles automatically, selecting the best based on a chosen metric (e.g., AUC-ROC). AutoML explicitly detects and handles class imbalance via oversampling/undersampling which is a direct advantage for fraud datasets. Custom training supports any Python framework via Script Run or

	YAML-defined component pipelines. Distributed training across compute clusters uses PyTorch DDP or Horovod.
Model Evaluation	AutoML generates a comprehensive evaluation dashboard including confusion matrix, calibration plots, AUC-ROC, lift chart, and feature importance. The Responsible AI Dashboard integrates fairness assessment, error analysis, and model explanations (SHAP, LIME) is critical for HIPAA-regulated healthcare applications. All runs are tracked in Azure ML Experiments for full reproducibility.
Deployment & Inference	Production endpoints are deployed to Azure Kubernetes Service (AKS) for scalable, managed real-time inference. Azure Container Instances (ACI) provides a lightweight option for development and testing. Batch endpoints process large claim files asynchronously with built-in parallelism. Azure API Management fronts the ML endpoint with authentication, rate limiting, and logging.
Monitoring	Azure ML Model Monitoring tracks data drift, prediction drift, and feature distribution changes on a scheduled basis. Azure Monitor and Application Insights provide full telemetry on endpoint latency and error rates. Azure ML Pipelines support automated retraining triggers based on monitoring alerts.

4. COMAPARITIVE ANALYSIS:

The three platforms are evaluated across seven dimensions most critical to a production healthcare fraud detection deployment. The assessment reflects both the general capabilities of each platform and their specific fit for the characteristics of this use case: regulated environment, class-imbalanced tabular data, and real-time inference requirements.

Dimension	AWS SageMaker	Google AutoML	Azure ML Studio
Ease of Use	Moderate & powerful but steep learning curve; SDK-heavy. Data Wrangler helps non-engineers.	High & AutoML Tables requires minimal ML expertise; intuitive automated end-to-end pipeline.	High because Designer drag-and-drop lowers barrier; AutoML wizard guides beginners step-by-step.
Flexibility	Very High & has full control over algorithms, containers, and frameworks (TF, PyTorch, custom Docker).	Moderate as AutoML is a black box; custom training available but less seamlessly integrated.	High for both visual Designer and code-first SDK; supports any Python framework via custom environments.

Scalability	Very High as it has distributed training, spot instances, serverless inference, multi-model endpoints.	High for serverless AutoML scales automatically; custom training on TPUs for large models.	High because AKS for production, auto-scaling compute clusters, distributed training supported.
Explainability / Compliance	High and SageMaker Clarify: bias detection + SHAP explanations; HIPAA-eligible.	Moderate-High as Vertex Explainable AI provides SHAP; HIPAA compliance requires additional BAA.	Very High because Responsible AI Dashboard is the most comprehensive fairness + explanation suite; HIPAA-eligible with BAA.
MLOps Maturity	Very High, SageMaker Pipelines, Model Registry, A/B testing, Model Monitor fully integrated.	High, Vertex AI Pipelines (Kubeflow) is powerful but requires more setup than SageMaker.	High, Azure ML Pipelines, model registry, and GitOps integration via Azure DevOps.
Pricing Model	Pay-per-use (compute, storage, endpoints). Costs can escalate with prolonged endpoint uptime.	AutoML training charged by node-hour; prediction by request. Transparent and competitive.	Pay-per-use; AutoML priced by compute time. Azure Hybrid Benefit reduces cost for existing customers.
Ecosystem Integration	Best for AWS-native stacks (Redshift, Glue, Lambda, SNS). Deep AWS ecosystem integration.	Best for GCP-native stacks (BigQuery, Dataflow, Looker). Excellent for data-warehouse-centric workflows.	Best for Microsoft-ecosystem orgs (Data Factory, Power BI, Synapse, Active Directory RBAC).

Table 1: Platform Comparison Across Key Dimensions

Strengths and Limitations Summary

AWS SageMaker	 Strengths	 Limitations
	• Most flexible and customizable platform	• Steepest learning curve of the three platforms
	• Best MLOps tooling (Pipelines, Experiments, Monitor)	• Fragmented UX across many separate services
	• Largest ecosystem of built-in algorithms	• Cost management requires careful planning

Google AutoML (Vertex AI)	✓ Strengths	⚠ Limitations
	• Easiest to use — minimal ML expertise needed	• AutoML is a black-box — limited model transparency
	• Excellent BigQuery and data warehouse integration	• Less mature MLOps pipeline vs. SageMaker
	• Strong auto-generated model evaluation reports	• Custom training less integrated than AutoML path

Azure ML Studio	✓ Strengths	⚠ Limitations
	• Best Responsible AI / fairness tooling for regulated industries	• Designer pipelines can feel less flexible than pure code
	• Excellent Microsoft-ecosystem integration	• AKS deployment adds operational complexity
	• Dual interface (visual Designer + SDK) suits mixed teams	• Feature Store still in preview; some features immature

★ Recommended Platform for This Case

Azure ML Studio is the strongest fit for healthcare insurance fraud detection. The Responsible AI Dashboard provides model explainability and fairness auditing capabilities that regulators and compliance officers expect in HIPAA-governed environments. The dual interface (visual Designer + Python SDK) accommodates data engineers, data scientists, and business analysts within the same workspace. Azure AutoML's built-in class imbalance handling directly addresses one of the most common pain points in fraud detection datasets.

Runner-up: AWS SageMaker is the better choice when the organization already operates on AWS, needs maximum flexibility in model architecture, or has a mature data engineering team comfortable with the AWS ecosystem and Python SDK.

5. CONCLUSION:

All three platforms, AWS SageMaker, Google AutoML (Vertex AI), and Azure ML Studio can successfully implement the healthcare insurance fraud detection ML pipeline from data ingestion to production monitoring. Each platform offers a complete pathway through the ML lifecycle, with meaningful differences in how that pathway is structured and who it is designed to serve.

- **Choose AWS SageMaker** when maximum customization, a large built-in algorithm library, and robust MLOps at scale are required and the engineering team has the depth to leverage the platform's full capabilities.

- **Choose Google AutoML (Vertex AI)** when speed-to-value and automation are priorities, and data already lives in BigQuery or GCS. It is the most accessible option for teams newer to ML or operating with limited data science resources.
- **Choose Azure ML Studio** when operating in a Microsoft ecosystem, when regulatory compliance and model explainability are non-negotiable, or when the team spans both technical and non-technical users.

The table below provides a final at-a-glance recommendation matrix for common organizational scenarios:

Organizational Scenario	Recommended Platform
Startup or small team with limited ML expertise	Google AutoML (Vertex AI)
Enterprise on AWS with custom model requirements	AWS SageMaker
Healthcare or finance with regulatory compliance focus	Azure ML Studio
Data-warehouse-centric analytics team on GCP	Google AutoML + BigQuery
Mixed team of engineers and business analysts	Azure ML Studio (Designer)
Research team requiring cutting-edge model architectures	AWS SageMaker (Script Mode)

Table 2: Platform Recommendation by Organizational Scenario

This exploratory exercise reinforces a key insight in ML engineering: there is no universally superior platform. The optimal choice depends on the intersection of organizational infrastructure, team expertise, regulatory environment, and the specific characteristics of the ML problem at hand. Developing the ability to evaluate these trade-offs critically is a core competency for any ML engineer operating in cloud environments.